



EE551 – Mini-Project

Marking Scheme and Other Guidelines



EE551 Mini-project Marking Scheme

100 Marks total

Parking Space Detection Algorithm and Results: 45 marks

- 10 images in total, three groups, getting progressively more difficult.
- Marks will be allocated on the basis of the effectiveness of the solution presented in terms of detecting parking spots (i.e. % detected, accuracy of localisation) as well as classification of occupancy (occupied, free etc.)
- More marks allocated for “harder” images

Discussion and Critique: 35 marks

- Discussion of results, what worked well, what didn't. Contribution of each block in your processing chain, How much did you have to change the model for each set of images (generalisation). Discussion of embedded considerations (real-time performance, storage and processing power).

Code: 5 marks

- Good coding practice, neatness, commentary etc.

Report Structure and Layout etc: 15 marks

- General quality of report as per guidelines given in presentation. Maximum of 4 pages. Approx. 2 pages for your methodology, approx. 2 pages for Discussion and Critique.
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Guidance: Classical vs Deep-Learning Approaches

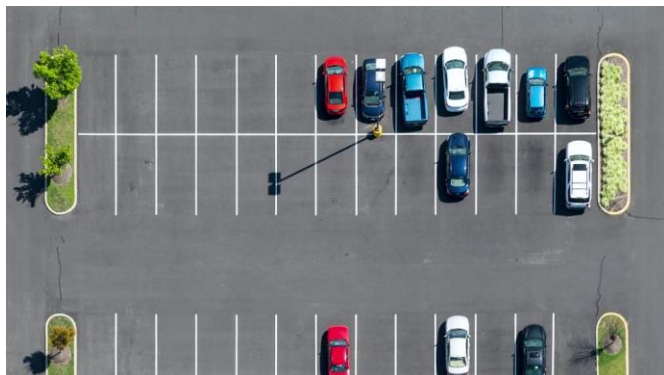
- The primary goal is to complete the task using what have been called “classical” image processing techniques in class, i.e. that have some sort of underlying analytical basis, and not just a machine learning/AI algorithm that is trained from data (i.e. "data-driven" or "black box"). This could be based on techniques presented in class, or suggested in the presentation that describes the mini-project.
 - You can use other techniques if you wish, and indeed, it’s a positive thing if you do that as it shows extra initiative, but these should also be “classical” approaches. Remember, the goal is for you to demonstrate that you understand the underlying techniques and can interpret the results, hence the focus on “low level” image processing operations.
 - However, you can, if you wish, compare your classical implementation against e.g. a pre-trained deep-learning model and provide a discussion in the analysis section. But **your primary solution should be based on classical approaches** - a submission that is based mainly on machine learning/AI approaches will not result in a good grade.
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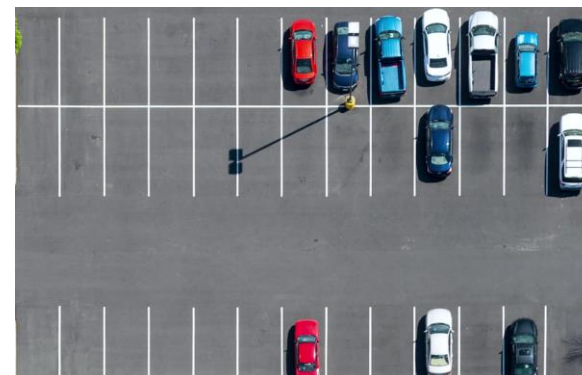
Guidance: Identification of Regions of Interest (ROIs)

- The goal should be to develop a fully-automated solution for all of the test sets, however, if you're having difficulty (especially with the “medium” and “hard” test images), you can **manually** define some ROIs within the image to allow you to continue making progress.
 - However, these ROIs shouldn't be the individual car parking spots, rather what you select should be the parking “zones” and you work from there. Examples of "acceptable" and "unacceptable" ROIs are given below.
 - So, you can use a manual approach as a “fall-back” solution if it allows you to complete the assignment, however, more marks will be awarded for more automated approaches.
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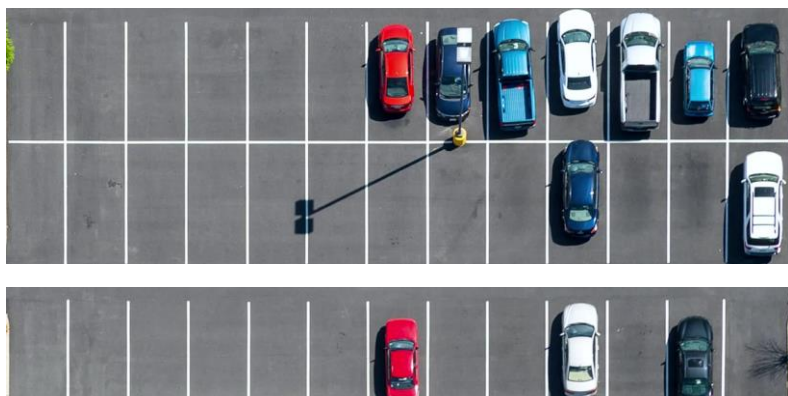
Examples of Manually-derived Parking Zones



Sample Input Image



Example 1: Manually-defined ROI of parking "zone" - acceptable



Example 2: Manually-defined ROI of parking "zone" - acceptable



Example 3: Manually-defined ROI of parking **space** – not acceptable!



Logistics

- Submission Deadline: **11:59 pm** on **Sunday 23rd November**
 - Like the assignments, the mini-project is an **individual** project
 - Please follow the guidelines for your mini-project submission carefully – see previous Presentation as well
 - Both a Jupyter Notebook (with your code) and a separate PDF submission are required
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